HYACDCSIM - User manual

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Steady-state model of an MTDC system in PSS/E

1.1 Introduction

This chapter describes the steady-state user-written model of a MTDC and its use within PSS/E.

1.2 Steady-state model

This section describes the steady-state model of an MTDC system developed in [2]. The MTDC system consists VSC interconnected by a DC grid. The ideas behind VSC modeling for power flow analysis are depicted in Fig. 1.1 [1].

Each converter is connected to the AC grid and to the DC grid. The AC-side of the VSC is modeled by a voltage source $\bar{e}_c = e_c \angle \delta_c$ coupled to the AC bus s ($\bar{u}_s = u_s \angle \delta_s$) through a phase reactor, a capacitor and a transformer ($\bar{z}_c = r_c + j \cdot \omega \cdot L_c$, $\bar{z}_f = -j1/(\omega \cdot C_f)$ and $\bar{z}_{tf} = r_{tf} + j \cdot \omega \cdot L_{tf}$, respectively). This LCL circuit serves as a filter to reduce the harmonics produced by the switching of the VSC.

The DC-side of the converter is modeled by a current source i_{dc}, injecting current into the DC grid. AC and DC sides are related by the energy conservation principle. The active power, p_c, injected into the AC grid is equal to the power absorbed from the DC grid, p_{dc}, and the converter losses. The converter losses, p_{loss}, can be calculated using a quadratic function of the converter AC current magnitude, i_c, as proposed in [3]:

$$p_c + p_{dc} + p_{loss} = 0, \qquad p_{loss} = a + b \cdot i_c + c \cdot i_c^2, \tag{1.1}$$

The converter is able to control (a) the active power injected into the AC bus, p_o, the frequency applied at the AC bus, or the DC voltage, u_{dc}, and (b) the reactive power injected into the AC bus, q_o, or the magnitude of the AC voltage, u_f. Note that these values are initialized according to the values initially imposed at bus s. The choice of the variables to be controlled also depends on whether the VSC operates in GFol or GFor mode. Commonly, the converter is operated in GFol mode. In steady state, the operation mode is not relevant. The VSC are assumed to be PV or PQ nodes for the AC power flow.

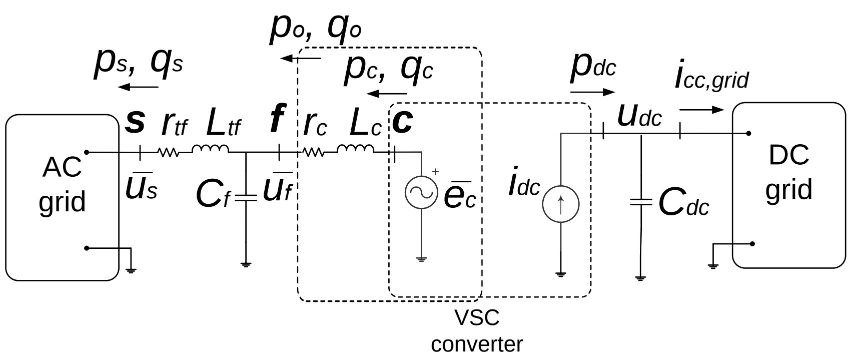


Figure 1.1: VSC converter modelling [1].

A general MTDC system with n converters and n_L DC lines is considered. For the power flow analysis, it is assumed that one converter controls the DC voltage (DC slack) and the remaining VSC control the active power. The sequential approach proposed in [4] has been used because it can be easily implemented in PSS/E without modifying the AC power-flow algorithm. Therefore, the use of MTDC technology can be easily explored using existing cases of large AC systems. The sequential approach essentially solves the AC and DC power flow in an iterative process. In the DC power flow, the DC slack controls the DC voltage, u_{dc}, and an extra iteration is needed to compute the active power specified by the AC power flow [4]. The

DC-slack fixes the DC voltage, u_{dc}, whereas the remainder nodes fix the active power, p_{dc}. The sequential approach can be summarized as follows.

1. An initial AC power flow analysis is performed (by calling PSS/E), assuming a direct connection of the VSC to bus s with the specified AC side settings to calculate and extract the unknown values. Next, the explicit transformer model is considered, and a virtual bus is introduced at point f, with the VSC connected to this bus. Finally, backward calculations are performed to determine the required settings at bus f according to the converter control (i.e., p_o and q_o or u_f).

2. The k-th external iteration starts with an initial guess of the active power injected into the AC grid by the VSC at the DC-slack bus ($p_{o,n_s}^{(k)}$, where n_s is the index of the DC-slack bus). If available, the results for iteration $k-1$ are used.

3. An AC power flow is calculated (by calling PSS/E) taking the variables of the current external iteration k as the initial state (or an initial guess, if started from scratch). All the VSCs are PQ or PV buses (at the virtual bus of the f-point in Fig. 1.1), depending whether they are controlling the reactive power or the AC voltage. The specified active power of the non-slack converters is constant during the external iteration, whereas the specified active power of the DC slack is allowed to vary during the external iteration ($p_{o,n_s}^{(k)}$). This is a traditional AC power flow with its own slack bus.

4. The AC/DC coupling for each converter is solved. The connection impedance is taken into account to calculate $p_{c,i}^{(k)}$ (at c-bus in Fig. 1.1) from the virtual bus of the f-point data. The converter losses are also calculated to obtain the power at the DC side of each converter station, for the inner DC power flow ($p_{dc,i}^{(k)}$).

5. The DC power flow (by using a Newton-Raphson algorithm) is solved. The DC slack bus specifies the fixed DC voltage (u_{dc,n_s}^{0}) and the rest of DC buses specify the active power injection ($p_{dc,i}^{(k)}$, which are data for the DC power flow, but are updated as described in the previous point). DC voltages are obtained together with the power of the DC slack ($p_{dc,n_s}^{(k)}$).

6. DC slack iteration (ℓ): A new value of $p_{o,n_s}^{(k+1)}$ is obtained, iteratively, from $p_{dc,n_s}^{(k)}$ taking into account the converter losses [4]:

   (a) Initial value: $p_{c,n_s}^{(\ell=0)} = p_{c,n_s}^{(k)}$.

   (b) Solve the branch $f, n_s - c, n_s$ (Fig. 1.1, for the DC-slack bus n_s) with the data: u_{o,n_s}, δ_{o,n_s}, q_{o,n_s} and $p_{c,n_s}^{(\ell)}$ (Newton-Raphson method).

   (c) Obtain the new value of $p_{c,n_s}^{(\ell+1)}$ with p_{dc,n_s} and $p_{loss,n_s}(i_{c,n_s}^{(\ell)})$ using (1.1).

   (d) If $|p_{c,n_s}^{(\ell+1)} - p_{c,n_s}^{(\ell)}| < \epsilon$, stop; if not, put $\ell = \ell + 1$ and return to step b). The output is $p_{o,n_s}^{(k+1)}$.

7. Convergence test: If $|p_{o,n_s}^{(k+1)} - p_{o,n_s}^{(k)}| < \epsilon$: stop and if not, put $k = k + 1$ and return to step 1).

Although the proposed formulation is described for a single MTDC system, multiple MTDC systems can also be accommodated, with their DC power flows solved simultaneously.

1.3 Implementation

The sequential approach has been completely implemented in Python 2.7 for PSS/E version 34. PSS/E is called at each external iteration to solve the DC power flow. The DC power flow and the DC-slack iteration have also been coded in Python and are called once the AC power flow has been solved. The final solution of the power flow can later be used by PSS/E as the initial operating point for dynamic simulation.

The following Python files are available:

- main_hyacdcsim.py is the main code that implements the building of the static and dynamic MTDC systems and the solutions of the resulting AC/DC power flow. This code makes use of a module called acdc_module.py.

- module_acdc.py is the module in charge to set up and solve the sequential AC/acdc power flow algorithm for MTDC systems. PSS/E solves the AC power flow, whereas a Python module, module_dclf.py, solves the DC power flow and the copupling between both the AC and DC grids. In addition, the functions of this module also show the solution of the AC/acdc power flow and write data files needed for dynamic simulations. The following functions are implemented:

– main_runacdclf: principal function to prepare and run the AC/DC power flow

– fun_readMTDCdata: extract the MTDC system data from a predefined Excel file

– fun_getT2Piequivalent: build π-equivalent of the transformer and LC filter

– fun_setinitialvaluesacdclf: initialize sequential AC/DC power flow

– fun_setACsetpoints: set AC-side generator set points

– fun_backwardcomputation: do a backward computation by explicitly representing the transformer and shunt branch, and then recompute the voltage and power set points at point f

– fun_addnewelements: Add a virtual bus, a shunt filter, and a transformer filter, which are connected between the AC bus and the virtual bus

– fun_relatedvirtualbuses: retrieve the virtual bus corresponding to a specific AC bus

– fun_getACresults: extract results from the AC power flow run

– fun_calculateVSCtransformerlosses: calculate the active and reactive power losses of VSC-connected transformers

– fun_initializeDCfromAC: initialize DC-side variables from AC results

– fun_calldclf: call the DC power flow (DC load flow and DC slack bus iteration)

– fun_getacdcoutput: get AC/DC power flow results

– fun_showacdclfoutput: show results

– fun_setacdcdynamicdata: create grid and element dynamic data

– fun_updatedynamicfile: update .dyr file with VSC models and DC grid model

– fun_setDCnetworkdata: set DC grid data for dynamic simulation

– fun_setMTDCdyrdata: set VSC model and DC grid model parameters

- fun_write2txtfile: write DC grid data to .txt file
- fun_setDCnetworkdata: organize and write MTDC grid data, including buses, branches, and grid parameters, to text files

- module_dclf.py contains those functions needed to prepare and solve the DC grid power flow. The following functions are implemented:
  - fun_mainDCloadflow: principal function to solve the DC power flow
  - fun_getDCjacobian : builds the DC grid Jacobian matrix
  - fun_getDCconductance: builds the DC grid admittance matrix
  - fun_getDCincidence: builds the DC grid incidence matrix
  - fun_solveDCloadflow: solves the DC power flow
  - fun_slackiterationDC: iteration of the DC slack bus converter

1.4 User manual

1.4.1 Requirements

PSS/E version 34 uses Python 2.7 (32 bit). A compatible Numpy work package is required, too. Type the following command in the command prompt when located at \\Python27\\Script:

pin install "name work package"

1.4.2 Organization and installation

The Hybrid AC/DC system simulator (HYACDCSIM) is organized around for folders:

- _ForUserModels: This folder includes the user-written dynamic models of the MTDC system. These models are described in chapter 2.
- _PyModules: The _PyModules folder contains the Python modules described in section 1.3.

- Input: This folder can contain subfolders, each containing the initial power flow (.sav) and dynamic data (.dyr) files of the case to be studied as well as the Python file to define the MTDC system.
- Simulation: The Simulation folder can contain subfolders, each containing the necessary files for the dynamic simulation of the use case to be studied. These files are:
  - MTDC.dll: the compiled dynamic link library (DLL) of the user models
  - data_Buses_base.txt: DC network ID, the bus numbers of the DC grid and their corresponding number in the AC grid, the initial voltage of the DC buses in p.u., and DC bus conductance and capacitor data
  - data_Lines_base.txt: DC network ID, and DC line resistance and inductance data
  - data_acdcbus.txt: the bus numbers of the DC grid and their corresponding number in the AC grid.
  - data_Ac.txt: the DC incidence matrix
  - data_Ydc: the DC admittance matrix.

  Note that the .txt files are automatically generated by HYACDCSIM.

The main_hyacdcsim.py file must be located at the same level as these four folders.

1.4.3 Defining MTDC systems

The steady-state data of the MTDC system is defined by means of an Excel file define_grids_mtdc.xlsx. In the main file, HYACDCSIM, the path to this file should be defined. The Excel file is structured such that each DC network is assigned a unique ID. The required data is organized into four sheets named *baseMVA*, *converter*, *dcbus*, and *dcbranch*.

The first sheet, *baseMVA*, contains the power rating of the MTDC system.

The second sheet, *converter*, is a table that stores the converters' data. Each table row contains the information of one converter at a specific bus of the DC grid. The parameters needed are shown in Table 1.1. The pu values are given on the converter rating *(rateA)*.

Parameter	**Description**	**Typical value**
MTDC_network_ids	DC network ID	-
DC bus number	DC bus number	-
AC bus number	AC bus number (bus s)	-
r_t	Transformer resistance (pu)	0.01
x_t	Transformer reactance (pu)	0.1
b_f	Filter capacitance (pu)	0
r_c	Reactor resistance (pu)	0.005
x_c	Reactor reactance (pu)	0.1
a	Constant loss term (pu)	-
b	Linear loss term (pu)	-
c_{rect}	Quadratic loss term in rectifier mode (pu)	-
c_{inv}	Quadratic loss term in inverter mode (pu)	-
Pmax	Maximum active power (MW)	-
Pmin	Minimum active power (MW)	-
Qmax	Maximum reactive power (Mvar)	-
Qmin	Minimum reactive power (Mvar)	-
rateA	Rating A (MVA)	-
rateB	Rating B (MVA)	-
rateC	Rating C (MVA)	-
ratio	Not used	0
angle	Not used	0
status	Not used	1
angmin	Not used	0
angmax	Not used	0

Table 1.1: VSC parameters

The third sheet, *dcbus*, is a table and stores DC bus data. Each table row contains the information of one DC bus. The parameters needed are shown in Table 1.2. The pu values are given on the rating of the MTDC system (*baseMVA*). A typical value for C_{dc} is $195 \cdot 10^{-6} F$.

Parameter	**Description**	**Typical value**
MTDC_network_ids	DC network ID	-
DC bus number	DC bus number	-
P control	1: P, 2: DC slack	-
Q control	1: q_s, 2: u_s	-
u_s	Voltage magnitude at bus s	1.0
δ_s	Voltage angle at bus s	0.0
P_s	Active power (MW)	-
Q_s	Reactive power (Mvar)	-
u_{dc}	DC voltage set point (pu)	1.0
P_{dc}	Power injected into the DC grid (MW)	-
i_{dc}	DC-side current of the VSC (pu)	-
g_{dc}	Bus conductance (pu)	0.0
C_{dc}	Bus capacitance (pu)	-
Area	Not used	0
baseKV	Not used	0
zone	Not used	0
Vmax	Not used	0
Vmin	Not used	0

Table 1.2: DC bus parameters.

The last sheet, *dcbranch*, is also a table and stores the data of the DC branches. Each table row contains the data of one DC branch. The parameters needed are shown in Table 1.3. The pu values are given on the rating of the MTDC system ($baseMVA$). Typical values for r_{dc}, L_{dc}, and C_{cc} are $0.02\Omega/km$, $10^{-3}H/km$, and $0.02 \cdot 10^{-6}F/km$.

Parameter	**Description**	**Typical value**
MTDC_network_ids	DC network ID	-
From DC bus number	DC bus number	-
To DC bus number	DC bus number	-
r_{dc}	Branch resistance (pu)	-
L_{dc}	Branch inductance (pu)	-
C_{cc}	Branch capacitance (pu)	-
rateA	Rating A (A)	-
rateB	Rating B (A)	-
rateC	Rating C (A)	-
ratio	Not used	0
angle	Not used	0
status	Not used	1

Table 1.3: DC branch parameters

Note that the dynamic model of the DC grid as implemented in HYACDCSIM makes use of C_{dc}, which aggregates the converter capacitance, C_{vsc}, and half of the capacitances of the branches connected to the VSC, C_{cc}.

1.4.4 Execution

Running the main code main_hyacdcsim.py creates the steady-state model of the MTDC system and solves the AC/DC power flow.

Before executing the main code, main_hyacdcsim.py needs to be modified according to the case under study. The necessary modifications are briefly indicated next:

- Lines 50 to 51 define the file names and paths of the power flow (.sav) and dynamic (.dyr) files of the initial case and need to be defined by the user.
- Line 52 defines the Excel file containting the MTDC system data.
- Lines 60 to 62 specify whether the dynamic model should be built, the AC/DC power flow results should be saved, and whether progress information should be shown to the user.
- Lines 65 and 66 specify power flow tolerances and maximum iteration numbers:

1.4.5 Model Output

A part from the optional .txt files for the dynamic simulations and the AC/DC power flow results, the execution of HYACDCSIM prints a summary of the results of the MTDC system power flow with respect to:

- DC bus quantities
- DC branch quantities
- losses of the AC/DC coupling (transformer, reactor, VSC)
- voltage, active and reactive power of the VSC

Power flow convergence is indicated, too.

1.4.6 Steady-State Case Study

In this section, the results obtained from the AC/DC power-flow calculation are illustrated using the Kundur two-area system as a test case. The original AC network is modified by introducing two electrically independent DC grids, each interfaced with the AC system through VSC converters. Their *baseMVA* is 100 MVA.

The first DC grid consists of three DC buses interconnected by three DC lines. Two VSC converters interface this DC grid with the AC system at buses 5 and 11. In the proposed PSS/E-based implementation, each VSC is represented on the AC side by an equivalent generator-type machine model together with its associated equivalent circuit. To accommodate this representation, two virtual AC buses, buses 12 and 13, are introduced. The machine models representing the corresponding converters are connected to these virtual buses, while the converter equivalent circuits connect to the respective physical AC buses 5 and 11.

The second DC grid consists of two DC buses interconnected by a single DC line and is interfaced with the AC system through two VSC converters associated with buses 6 and 10. Following the same modeling approach, two additional virtual AC buses, buses 14 and 15, are introduced. The equivalent machine models representing these converters are connected to the virtual buses, whereas their associated equivalent circuits connect to the physical AC buses 6 and 10.

It should be emphasized that the introduction of these virtual AC buses is a feature of the proposed VSC representation developed within the PSS/E environment. The virtual buses are introduced solely to facilitate the generator-type machine representation of the converters and do not correspond to physical buses of the DC grids. Physically, each VSC provides an interface between its corresponding AC bus and DC bus, whereas, in the proposed PSS/E implementation, its AC-side behavior is represented through the associated equivalent circuit and machine model connected to the corresponding virtual AC bus.

Before executing the program and extracting the results, certain parameters from Tables 1.1, 1.2, and 1.3 must be initialized. In this example, it is assumed that the following inputs are applied for VSCs:

Parameter	MTDC 1		MTDC 2	
	VSC 1	VSC 2	VSC 1	VSC 2
MTDC network ID	1	1	2	2
DC bus number	1	3	1	2
AC bus number (bus s)	5	11	6	10
Virtual AC bus number	12	13	14	15
Pmax	9999 MW	9999 MW	9999 MW	9999 MW
Pmin	-9999 MW	-9999 MW	-9999 MW	-9999 MW
Qmax	9999 Mvar	9999 Mvar	9999 Mvar	9999 Mvar
Qmin	-9999 Mvar	-9999 Mvar	-9999 Mvar	-9999 Mvar
rateA	500 MVA	500 MVA	500 MVA	500 MVA

Table 1.4: VSC parameters of the two MTDC systems considered in the case study.

Parameter	MTDC 1			MTDC 2	
	DC bus 1	DC bus 2	DC bus 3	DC bus 1	DC bus 2
MTDC network ID	1	1	1	2	2
P control mode	2	1	1	2	1
Q control mode	1	1	1	1	1
P_s	-200	0	200	-200	200
Q_s	0	0	0	0	0

Table 1.5: DC bus parameters of the two MTDC systems considered in the case study.

Parameter	MTDC 1			MTDC 2
	DC branch 1	DC branch 2	DC branch 3	DC branch 1
MTDC network ID	1	1	1	2
From DC bus number	1	1	2	4
To DC bus number	3	2	3	5

Table 1.6: DC branch parameters of the two MTDC systems considered in the case study.

The other parameters are used as presented in the aforementioned tables.

```
Converged after 2 iterations.
=====================================================================================================
DC grid - dcbus
=====================================================================================================
-----------------------------------------------------------------------------------------------------
| dcbus | dctype1 | dctype2 | us (pu) | deltas (deg) | Ps (MW) | Qs (Mvar) |  udc (pu) | Pdc (MW) |
-----------------------------------------------------------------------------------------------------
[[   1.           2.           1.           1.01729608   -6.26570306 -203.15223838   -0.00000053    1.            2.01955805]
 [   2.           1.           1.           1.            0.            0.            0.            0.99868459   -0.         ]
 [   3.           1.           1.           0.99991411   -6.69621501  200.07430273    0.00000026    0.99737017   -2.01275286]
 [   4.           2.           1.           1.00203037  -13.04810994 -203.51086812    0.00000023    1.            2.02273325]
 [   5.           1.           1.           0.96842921  -20.27410174  200.09408969   -0.00000021    0.99604856   -2.01374645]]
=====================================================================================================
DC grid - dcbranch
=====================================================================================================
-----------------------------------------------------------------------------------------------------
| dc From bus | dc To bus  | Pdc,ij (MW)| Pdc,ji (MW) | Icc,ij (pu)
-----------------------------------------------------------------------------------------------------
[[   1.           3.          134.58723071 -134.23328857    1.34587231]
 [   1.           2.           67.31858247  -67.23003126    0.67318582]
 [   2.           3.           67.18016272  -67.09174283    0.67268648]
 [   4.           5.          202.22333868 -201.42426447    2.02223339]]
=====================================================================================================
VSC converters - losses (MW) and AC/DC coupling
=====================================================================================================
-----------------------------------------------------------------------------------------------------
| transformer | reactor  | converter| total -Coupling AC/DC|
-----------------------------------------------------------------------------------------------------
[[0.79768902 0.39884451 0.         1.19653353]
 [0.80049653 0.40024826 0.         1.20074479]
 [0.82512188 0.41256094 0.         1.23768282]
 [0.85347416 0.42673708 0.         1.28021125]]
=====================================================================================================
VSC converters - Internal voltage
=====================================================================================================
-----------------------------------------------------------------------------------------------------
| ec (p.u) | deltac (deg) | Pc (MW) | Qc (Mvar) |
-----------------------------------------------------------------------------------------------------
[[   1.01438941  -10.78197667 -201.95580472   15.95277018]
 [   1.00924295   -2.14886137  201.27529404   16.01239468]
 [   0.99915818  -17.71155562 -202.27332454   16.5010329 ]
 [   0.97831771  -15.42969079  201.37465883   17.07306101]]
=====================================================================================================
MisMatches
=====================================================================================================
```

Figure 1.2: Simulation results of AC/DC power flow, with a focus on the DC section.

A summary of the results obtained from the AC/DC power-flow analysis for the case study is presented in Figure 1.2. The power-flow solution converges after two iterations. Since the modified Kundur system contains two independent DC grids, the reported results include the quantities associated with both DC networks as well as the four VSC converters interfacing them with the AC system. The results are organized into four sections.

- The first section presents the results associated with the DC buses. In total, five DC buses are considered: three buses belonging to the first MTDC network and two buses belonging to the second MTDC network. For each DC bus, the corresponding control types are reported together with the voltage magnitude and angle at the associated AC-side bus, the active and reactive powers (P_s and Q_s), the DC voltage magnitude (u_{dc}), and the power exchanged with the DC network (P_{dc}).

- The second section reports the DC branch quantities. The first MTDC network contains three DC branches connecting buses 1–3, 1–2, and 2–3, whereas the second MTDC network contains a single branch connecting buses 4 and 5. For each branch, the power flow at both ends of the line and the corresponding current magnitude in per unit are provided.

- The third section presents the losses associated with the four VSC converter interfaces. The losses of the transformer, reactor, and converter are reported separately, together with the total losses associated with the AC/DC coupling.

- The final section provides the internal electrical quantities of the four converters, including the magnitude and phase angle of the internal converter voltage as well as the corresponding active and reactive powers.

The DC-bus and DC-branch quantities are therefore reported for each of the two MTDC networks, whereas the converter-related quantities are provided individually for each of the four VSC converters.

Chapter 2

Dynamic model of an MTDC system in PSS/E

2.1 Introduction

This chapter describes the dynamic user-written model of a MTDC and its use within PSS/E. Dynamic simulations within PSS/E cover electromechanical dynamics, with time constants from 0.01 s to 10 s [5]. Electromechanical models of a power system take into account the slow dynamics of synchronous machines, their controllers, and other devices, whereas the AC branches are assumed to be quasi-static.

2.2 Dynamic model of an MTDC system

The model is split into converter models, the DC grid, and supplementary control models. Converter models are separated into GFol and GFor VSC models. Similar to the quasi-static AC branch modeling, the transformer, the capacitor, and the phase reactor of each VSC model shown in Fig. 1.1 are represented by a quasi-static model using algebraic relationships.

2.2.1 Dynamic model of the GFol VSC

The GFol VSC is controlled using vector control where the AC voltage at the capacitor is aligned with the d-axis: $\bar{u}_f = u_f + j0$. So far, no phase-locked loop (PLL) is implemented for this purpose. The active and reactive-power injections of the VSC at the capacitor are therefore $p_o = u_f \cdot i_{c,d}$ and $q_o = -u_f \cdot i_{c,q}$.

GFol VSCs are essentially current-controlled voltage sources. Active and reactive power are controlled with an inner current loop and with an outer controller [1]. The inner loop dynamics (response times from 1 to 10 ms [6]) are much faster than the ones of the synchronous generators and their controllers and they have been approximated by a first-order systems, as shown in Fig. 2.1, where $i_{c,d}^{ref}$ and $i_{c,q}^{ref}$ are the current references. The GFol VSC becomes thus a current source. Either the active power p_f or the DC voltage u_{dc} is controlled with the d-axis current $i_{c,d}$ by using PI controllers, as depicted in Figure 2.2a. Similarly, the q-axis current is used to control either the reactive power or the AC-voltage modulus (Fig. 2.2b). Note that the integrator state variables are initialized to 0. The time constants of the outer loops are between 1 ms and 100 ms [6].

The converter model was implemented with the operating limits for active and reactive power: P_{max}, P_{min}, Q_{max}, Q_{min} and with the maximum current limit $i_{c,max}$, which can be set to d-axis priority, q-axis priority or equal priority [7]. The maximum output AC voltage of a VSC depends on the DC voltage and the maximum modulation index ($e_{c,max} = m_{max} \cdot u_{dc}$) [8] and this limit is also taken into account: if $e_c^{ref} > e_{c,max}$, the current references are re-calculated using $e_{c,max}\angle\delta_c^{ref}$ as internal voltage, as depicted in Fig. 2.1. Note that since the dynamic model is a fundamental frequency model, the non-linear behavior of the modulated AC voltage can be assumed to be appropriately compensated.

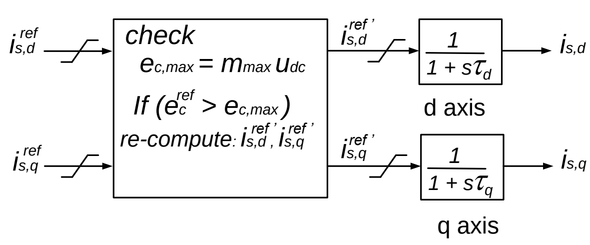
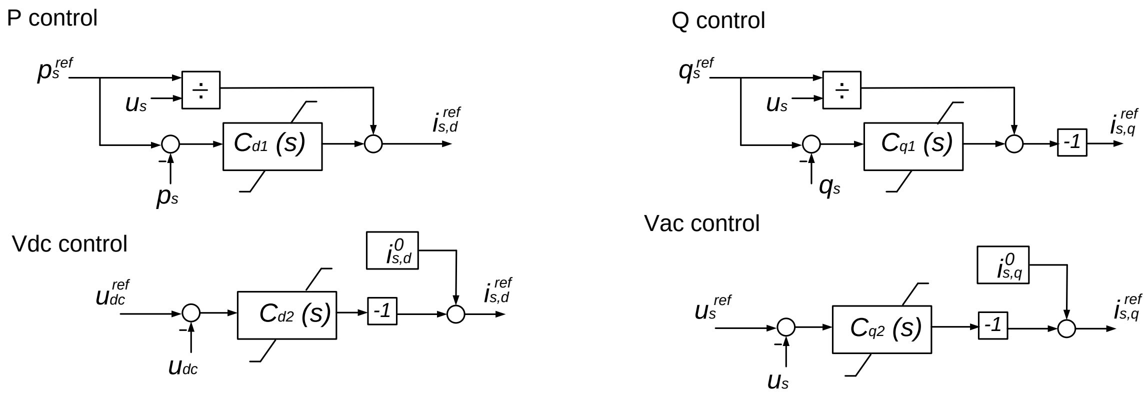


Figure 2.1: Approximation of the inner current loop.



(a) d-axis.

(b) q-axis.

$C_x(s) = K_{p,x} + K_{i,x}/s$ where x can be $d1$, $d2$, $q1$ or $q2$

Figure 2.2: Outer controllers.

2.2.2 Dynamic model of the GFor VSC

The GFor VSC is essentially implemented as current-limited voltage source. GFor VSC includes both inner voltage and current-control loops, where the voltage control usually acts on the AC voltage at the capacitor $\bar{u}_f$ in Fig. 1.1.

Given the time frame of interest, the voltage-control loop dynamics are approximated by a first-order system. The VSC GFor synchronizes with the AC voltage at the capacitor such that $\bar{u}_f = u_f + j0$. Different power-synchronizing mechanisms exist, all imitating to a certain extent the behavior of synchronous machines, some including PLL.

Whereas the q-axis voltage reference of the voltage source is set to 0 under no-fault operation, the d-axis voltage reference can be either fixed or made dependent on the reactive power of the VSC by means of a droop control. The droop control determined the d-axis reference voltage, $u_{f,d}^{ref}$, is:

$$\frac{u_{f,d}^{ref}}{dt} = \frac{q_o^{ref} - q_o(t)}{m_q \cdot T_q} - \frac{u_{f,d}^{ref} - u_f^{ref}}{T_q} \tag{2.1}$$

where q_o^{ref} and u_o^{ref} is the reference reactive power and the reference AC voltage, and m_q and T_q the droop and filter time constant. The power synchronization is formulated by now as a droop control as follows:

$$\begin{aligned} \frac{d\delta}{dt} &= \Omega_b \cdot (\omega - 1) \\ \frac{d\omega}{dt} &= \frac{p_o^{ref} - p_o(t)}{m_p \cdot T_p} - \frac{\omega - 1}{T_p} \end{aligned} \tag{2.2}$$

where δ and ω are the angle and angular speed of $\bar{u}_f$, p_f^{ref} is the reference active power, and m_p and T_p the droop and filter time constant.

The model includes additional active and reactive power limiter control not shown. These limiter controls guarantee limiting the steady-state current to $i_{c,max}$. In addition, a hard transient current limit is implemented, too. The transient limit must be above $i_{c,max}$ to avoid instability.

2.2.3 DC-grid model and AC/DC-coupling

The dynamic model of the DC grid includes the converters, the capacitors and the cables, as in [1]. The inputs of the system are the current injections of the VSC, $\boldsymbol{I_{dc}} = (i_{dc,i}) \in \mathbb{R}^{n\times 1}$. DC cables are represented by an equivalent π-model, with resistance $r_{dc,ij}$, inductance $L_{dc,ij}$ and capacitance $C_{cc,ij}$ (Fig. 2.3). An equivalent capacitor at each DC bus is used to model the capacitor of the converter, $C_{VSC,i}$, and the capacitance of the DC branches:

$$C_{dc,i} = C_{VSC,i} + \sum_{j\neq i} \frac{C_{cc,ij}}{2} \tag{2.3}$$

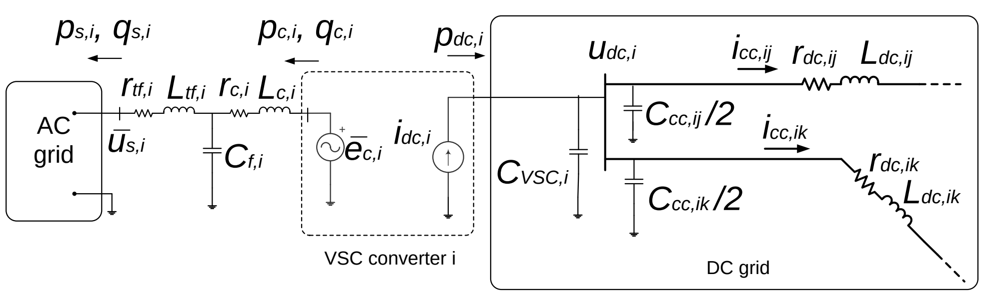


Figure 2.3: Dynamic model of the DC grid.

The model also includes a shunt conductance $g_{dc,i}$ at every DC bus that can be used to model resistive loads. The state variables of the system are the DC voltages, $\boldsymbol{U_{dc}} = (u_{dc,i}) \in \mathbb{R}^{n\times 1}$, and the currents through the DC lines, $\boldsymbol{I_{cc}} = (i_{cc,\ell}) \in \mathbb{R}^{n_L\times 1}$.

Therefore, the differential equations for the DC grid are [1]:

$$\boldsymbol{C_{dc}}\frac{d\boldsymbol{U_{dc}}}{dt} = -\boldsymbol{G_{dc}}\boldsymbol{U_{dc}} - \boldsymbol{A_c}\boldsymbol{I_{cc}} + \boldsymbol{I_{dc}} \tag{2.4}$$

$$\boldsymbol{L_{dc}}\frac{d\boldsymbol{I_{cc}}}{dt} = \boldsymbol{A_c^T}\boldsymbol{U_{dc}} - \boldsymbol{R_{dc}}\boldsymbol{I_{cc}} \tag{2.5}$$

where:

$$\boldsymbol{G_{dc}} = \mathrm{diag}(g_{dc,i}) \in \mathbb{R}^{n\times n} \tag{2.6}$$

$$\boldsymbol{C_{dc}} = \mathrm{diag}(C_{dc,i}) \in \mathbb{R}^{n\times n} \tag{2.7}$$

$$\boldsymbol{R_{dc}} = \mathrm{diag}(r_{dc,\ell}) \in \mathbb{R}^{n_L\times n_L} \tag{2.8}$$

$$\boldsymbol{L_{dc}} = \mathrm{diag}(L_{dc,\ell}) \in \mathbb{R}^{n_L\times n_L} \tag{2.9}$$

and $\boldsymbol{A_c} = (a_{i\ell}) \in \mathbb{R}^{n\times n_L}$ is the incidence matrix of the DC grid, whose elements are:

$$a_{i\ell} = \begin{cases} +1 & \text{if line } \ell \text{ is defined leaving node } i. \\ -1 & \text{if line } \ell \text{ is defined entering node } i. \\ 0 & \text{if line } \ell \text{ is not connected to node } i. \end{cases} \tag{2.10}$$

Finally, AC and the DC sides of each converter i are coupled by the energy conservation principle (1.1). At each time step, AC and the DC systems are updated sequentially and $p_{dc,i}$ is obtained from $p_{c,i}$. Currents injected into the DC grid are calculated as:

$$i_{dc,i} = \frac{p_{dc,i}}{u_{dc,i}} = \frac{-(p_{c,i} + p_{loss,i})}{u_{dc,i}} \quad \forall i = 1, \ldots, n \tag{2.11}$$

The static model of the DC grid can be easily derived from (2.4) and (2.5).

2.2.4 Active-power related supplementary controls of MTDC systems

The supplementary active power control includes distributed DC voltage control and frequency controls including droop control, time optimal control Lyapunov function and synthetic inertia.

DC voltage control

The most common options for the DC voltage control are: (a) a centralized control, in which one converter controls the DC voltage and the rest control the active power injected into the AC grid and (b) a distributed control, in which the DC voltage control is shared among a set of converters with the so-called DC voltage droop. A centralized control scheme based on GFol VSC is shown in section 2.2.1. In the distributed control scheme, the active-power set point of each VSC is given by [8]:

$$p_{o,i}^{ref}(t) = p_{o,i}^{0} - \frac{1}{k_{dc,i}}(u_{dc,i}^{0} - u_{dc,i}(t)) \tag{2.12}$$

Fig. 2.4 shows the distributed DC voltage control acting on the reference of the active-power outer controller (see Fig. 2.2).

Frequency control and synthetic inertia

Frequency control strategies can be divided into local and global strategies according to the setting of ω^*. The local strategy sets ω^* to its nominal value (i.e., 1 pu). The global strategy uses a weighted average of the frequencies measured at the AC side of the VSC stations of the MTDC system. The idea behind this control strategy is to inject active power from the AC terminals of the VSC where the frequency is above (below) the weighted average. Fig. 2.4 shows the implemented frequency control. Fig. 2.4 does not show the synthetic inertia, which modifies p_o^{ref} as follows:

$$p_o^{ref} = p_o^0 + \Delta p_o^{ref,DC} + \Delta p_o^{ref} - K_\alpha \cdot \frac{d\omega}{dt} \tag{2.13}$$

2.2.5 Reactive-power related supplementary controls of MTDC systems

The supplementary reactive power control includes AC voltage control and frequency-based power-oscillating damper (POD).Fig. 2.5 shows the implemented frequency-based POD. Fig. 2.5 does not show the AC voltage control, which modifies q_o^{ref} as follows:

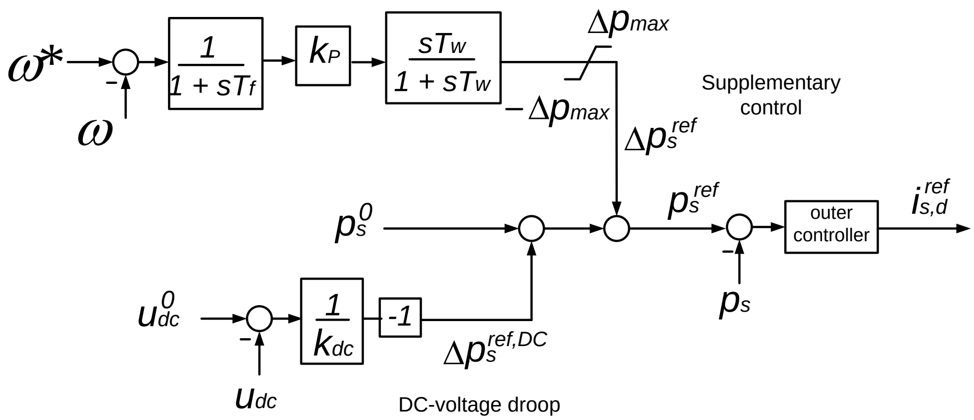


Figure 2.4: Active-power related supplementary controls of GFol-based MTDC systems.

$$q_o^{ref} = q_o^0 + \Delta q_o^{ref} + K_{ac} \cdot (u_f^{ref} - u_f) \tag{2.14}$$

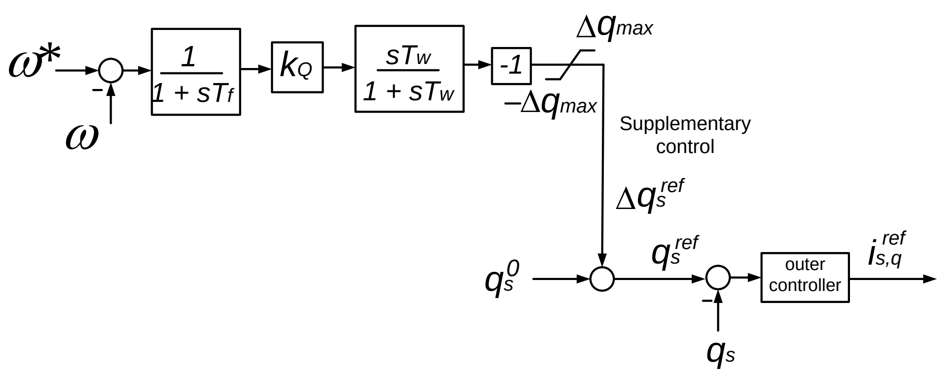


Figure 2.5: Reactive-power related supplementary controls of GFol-based MTDC systems.

2.3 Implementation

The dynamic model has been programmed (in FORTRAN) as a PSS/E user-defined model and consists of n_{vsc} "generator-type" models for the converters and one "governor-type" model for the DC grid.

DC-voltage droop control or controllers for ancillary services could have been included in the VSC models but, for flexibility, they were implemented in independent models which change the references of the converter models. Active power related controls have been implemented as "exciter-type" models, whereas reactive power related controls have been implemented as "stabilizer-type" models.

The active power related controls include the DC-voltage control and the frequency control that change the active-power set point of the converters (see for instance 2.12). The reactive power related controls include the AC voltage control and the POD modulating reactive-power set point of the converters in function of the frequency [9].

2.4 User manual

2.4.1 General Requirements

HYACDCSIM automatically creates and places the necessary .dyr and .txt files in the Simulation folder (see section 1.4. Note that the .dyr file needs to be adapted to set appropriate parameter values (see section 2.4.2). The DLL file and the .txt files need to be located in the same folder. An Intel Visual Fortran compiler compatible with PSS/E 34 is required to create the DLL file.

2.4.2 Parameterization of user-written models

The .dyr file sets the parameters of the dynamic models. The parameterization of user-written models follows a specific format according to the model type and the order of the list of parameters needed.

VSCGFL - VSC model

The VSC model is implemented as a generator-type model and computes the currents injected into the AC and the power injected into the DC grid. The converter is based on a grid-following control structure, in which the current dynamics are represented by a first-order system. The control is formulated in a (dq) reference frame aligned with the terminal voltage, and the current references are derived from the active- and reactive-power references. No PLL is explicitly represented in the current implementation. The VSC converter model, VSCGFL is implemented as a coordinated-call, current-injecting generator model (IC = 1 and IT = 1). The coordinated-call implementation is required due to the representation of the converter at the network interface by a Norton equivalent.

The following format is used in the .dyr file to assign and parameterize the VSCGFL model: *BUS 'USRMDL' IM 'VSCGFL' IC IT NI NC NS NV ICON(M) to ICON(M+5) CON(J) to CON(J+23)*

where

- BUS: the number of AC bus number f of the VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14')
- IC: 1
- IT: 1
- NI: 6
- NC: 24
- NS: 7
- NV: 29

The ICON and CON parameters are as follows:

Parameter	Description	Typical value
τ	Current loop time constant (s)	0.005
$K_{p,d1}$	P gain of P control (pu)	-
$K_{i,d1}$	I gain of P control ($pu \cdot s$)	-
$K_{p,d2}$	P gain of u_{DC} control (pu)	20.0
$Kd_{i,d2}$	I gain of u_{DC} control (pu)	400.0
$Kd_{d,d2}$	D gain of u_{DC} control (pu)	-
$K_{p,q1}$	P gain of Q control (pu)	-
$K_{i,d1}$	I gain of Q control (pu)	-
$K_{p,q2}$	P gain of u_f control (pu)	-
$K_{i,q2}$	I gain of u_f control (pu)	-
i_{max}	Max. current (pu)	1.1
$P_{o,max}$	Max. active power (MW)	-
$P_{o,min}$	Min. active power (MW)	-
$Q_{o,max}$	Max. reactive power (MVar)	-
$Q_{o,min}$	Min. reactive power (MVar)	-
$u_{dc,max}$	Max. u_{DC} (pu)	1.1
$u_{dc,min}$	Max. u_{DC} (pu)	0.9
m_{max}	Max. modulation index	1.3
a	Constant loss term (pu)	-
b	Linear loss term (pu)	-
c_{rect}	Quadratic loss term in rectifier mode (pu)	-
c_{inv}	Quadratic loss term in inverter mode (pu)	-
$T_{udc,min}$	Delay of the undervoltage protection (s)	0.1
$T_{udc,max}$	Delay of the overvoltage protection (s)	0.1

Table 2.1: CON parameters

Parameter	Description	Typical value
DCONTROL	P_o control: 1, U_{dc} control: 2	-
QCONTROL	Q_o control: 1, U_f control: 2	-
ILIMIT_PRIORITY	P-priority: 1, Q-priority: 2, Proportional: else	1
NDCBUS	Number of DC buses in the MTDC	-
NDCLINES	Number of DC lines in the MTDC	-
IDGRID	ID of the ith MTDC	-

Table 2.2: ICON parameters

The state variables are the following:

State	Description
$i_{c,d}$	d-axis current (pu)
$i_{c,q}$	q-axis current (pu)
x_{d1}	P control PI state (pu)
x_{d2}	u_{dc} control PI state (pu)
x_{q1}	Q control PI state (pu)
x_{q2}	u_f control PI state (pu)
x_{q2}	δ_f control PI state for passive grids (pu)

Table 2.3: STATE variables

VSCDRO - VSC model

The VSC model is implemented as a generator-type model and computes the voltage applied to the DC grid. The converter is based on a grid-forming control structure. The voltage dynamics are represented by a simple first-order system. The control is formulated in a dq reference system aligned with the voltage at bus f. No PLL is used. The droop implementation controls the angle/frequency and voltage. The VSC converter model, VSCDRO is implemented as a coordinated-call, current-injecting generator model (IC = 1 and IT = 1). The model is similar to the REGFM_A1 model. The coordinated-call implementation is required due to the representation of the converter at the network interface by a Norton equivalent.

The following format is used in the .dyr file to assign and parameterize the VSCDRO model: *BUS 'USRMDL' IM 'VSCDRO' IC IT NI NC NS NV ICON(M) to ICON(M+4) CON(J) to CON(J+21)*

where

- BUS: the number of AC bus number f of the VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14')
- IC: 1
- IT: 1
- NI: 5
- NC: 22
- NS: 9
- NV: 17

Note that ZSORCE is either the transformer impedance or very small (0.001 pu) if a transformer is explicitly modeled.

The ICON and CON parameters are as follows:

Parameter	**Description**	**Typical value**
τ	Current loop time constant (s)	0.01
m_p	Active-power droop gain (pu)	0.02
T_p	Active-power droop time constant	0.1
m_q	Reactive-power droop gain (pu)	0.05
T_q	Reactive-power droop time constant	0.10
$K_{p,dc}$	P gain of u_{DC} control (pu)	-
$K_{i,dc}$	I gain of u_{DC} control (pu)	-
$K_{p,p,max}$	P gain of active-power limiter (pu)	-
$K_{p,i,max}$	I gain of active-power limiter (pu)	-
$P_{o,max}$	Max. active power (MW)	-
$P_{o,min}$	Min. active power (MW)	-
$K_{q,p,max}$	P gain of reactive-power limiter (pu)	-
$K_{q,i,max}$	I gain of reactive-power limiter (pu)	-
$Q_{o,max}$	Max. reactive power (MVar)	-
$Q_{o,min}$	Min. reactive power (MVar)	-
a	Constant loss term (pu)	-
b	Linear loss term (pu)	-
c_{rect}	Quadratic loss term in rectifier mode (pu)	-
c_{inv}	Quadratic loss term in inverter mode (pu)	-
i_{max}	Max. current (pu)	1.0
$e_{c,max}$	Max. voltage	-
$e_{c,min}$	Min. voltage	-

Table 2.4: CON parameters

Parameter	Description	Typical value
ISHARDCLIMIT	Hard current limit: 0 or 1	-
ILIMIT_PRIORITY	P-priority: 1, Q-priority: 2, Proportional: else	1
NDCBUS	Number of DC buses in the MTDC	-
NDCLINES	Number of DC lines in the MTDC	-
IDGRID	ID of the ith MTDC	-

Table 2.5: ICON parameters

The state variables are the following:

State	Description
x_{ec}	AC voltage first-order dynamics' state (pu)
δ	angle (pu)
x_{dc}	u_{dc} control PI state (pu)
x_p	Active-power control PI state (pu)
x_q	Reactive-power control PI state (pu)
x_{pmax1}	Active-power limit PI state (pu)
x_{pmax2}	Active-power limit PI state (pu)
x_{qmax1}	Reactive-power limit PI state (pu)
x_{qmax2}	Reactive-power limit PI state (pu)

Table 2.6: STATE variables

XDCGRD - DC grid model

The DC grid is implemented as a governor-type model. Two alternative representations of the DC grid are implemented: a dynamic model, denoted as DDCGRD, and a static model, denoted as SDCGRD. Both models represent an MTDC network composed of n_b DC buses and n_L DC lines, but they differ in the formulation used to solve the DC network.

The DDCGRD model represents the DC grid by means of differential equations and therefore explicitly includes the dynamic behavior of the network. The DC lines are represented by a π-equivalent model. The series branch consists of the line resistance and inductance, while the shunt capacitances at the line ends are aggregated at the corresponding DC buses. Accordingly, the bus voltages and line currents constitute the dynamic states of the network. For a DC grid consisting of n_b buses and n_L lines, the resulting formulation contains $(n_b + n_L)$ state variables.

In the case of multiple MTDC systems, the implementation must account for the fact that a model instance cannot be called with different numbers of state and algebraic variables. Therefore, the required number of state variables is determined when the AC/DC system model is constructed.

As an alternative, the SDCGRD model provides a static representation of the DC network. In this formulation, the DC grid is solved using power-flow equations rather than differential equations. Consequently, no state variables are introduced for the DC network. This representation is suitable when the dynamic behavior of the DC grid is neglected and only the instantaneous steady-state relationship between DC bus voltages, line flows, and converter power injections is required.

The following formats are used in the `.dyr` file to assign and parameterize the two DC grid models:

BUS 'USRMDL' IM 'DDCGRD' IC IT NI NC NS NV ICON(M) to ICON(M+3)

BUS 'USRMDL' IM 'SDCGRD' IC IT NI NC NS NV ICON(M) to ICON(M+4)

where

- BUS: the number of the AC bus f of the first VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14')
- IC: 5
- IT: 0
- NI: 4 for DDCGRD and 5 for SDCGRD

- NC: 0
- NS: $n_b + n_L$ for DDCGRD and 0 for SDCGRD
- NV: $2 \cdot n_b$ for DDCGRD and $(n_b + n_L + 2 \cdot n_b)$ for SDCGRD

The ICON parameters are as follows:

Parameter	**Description**	**Typical value**
n_b	Number of DC buses of the ith MTDC	-
n_L	Number of DC lines of the ith MTDC	-
NPOLES	Number of poles of each converter	-
IDGRID	ID of the ith MTDC	-
MAX_ITER	Max. iteration number for static solution	-

Table 2.7: ICON parameters

The state variables of the dynamic DC grid model, DDCGRD, are the following:

State	**Description**
$\boldsymbol{U_{dc}}$	DC bus voltages ($\in \mathbb{R}^{n_b \times 1}$)
$\boldsymbol{I_{cc}}$	DC line currents ($\in \mathbb{R}^{n_L \times 1}$)

Table 2.8: STATE variables

SPWDRD - Active power related control

The supplementary active power control is implemented as an excitation-type model. The supplementary active power control includes distributed DC voltage control and frequency controls such as droop control, time optimal control Lyapunov function, and synthetic inertia.

The following format is used in the .dyr file to assign and parameterize the SPWDRD model: *BUS 'USRMDL' IM 'SPWDRD' IC IT NI NC NS NV ICON(M) to ICON(M+2) CON(J) to CON(J+11)* where

- BUS: the number of the AC bus f of the VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14')
- IC: 4
- IT: 0
- NI: 3
- NC: 12
- NS: 2
- NV: 15

The ICON and CON parameters are as follows:

Parameter	Description	Typical value
T_f	Frequency filter time constant (s)	0.1
T_w	Wash-out time constant (s)	10.0
K_{dc}	DC voltage droop (pu)	0.1
K_f	Frequency control gain (pu)	200.0
K_H	Synthetic inertia gain (s)	10
$u_{dc,thr}$	DC voltage threshold (pu)	10^{-6}
f_{thr}	Frequency threshold (pu)	10^{-6}
$\dot{f}_{thr}$	rate-of-change of frequency (ROCOF) threshold (pu/s)	10^{-6}
ΔP_{max}	Max. power variation (pu)	1.0
ΔP_{min}	Min. power variation (pu)	-1.0
$Ramp_p$	Max. power ramp (pu/s)	1000
B_{tocl}	Time optimal control gain	0.2

Table 2.9: CON parameters

Parameter	Description
ISDCCNTRL	Enable distributed DC control (0/1)
ISFCNTRL	Frequency control: 1, time-optimal control Lyapunov function: 2, AC-line emulation: 3, disabled: 0
ISALPHACNTRL	Enable synthetic inertia (0/1)

Table 2.10: ICON parameters

The state variables are the following:

State	Description
x_{ω}	Frequency lag filter
x_w	Wash-out filter

Table 2.11: STATE variables

SQWDRD - Reactive power related control

The supplementary reactive power control is implemented as a stabilizer-type model. The supplementary reactive power control includes AC voltage control and frequency-based POD control.

The following format is used in the .dyr file to assign and parameterize the SPWDRD model:

BUS 'USRMDL' IM 'SQWDRD' IC IT NI NC NS NV ICON(M) ICON(M+1) CON(J) to CON(J+9)

where

- BUS: the AC bus number of the VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14').
- IC: 3
- IT: 0
- NI: 2
- NC: 10
- NS: 2
- NV: 10

The ICON and CON parameters are as follows:

Parameter	Description	Typical value
T_f	Frequency filter time constant (s)	0.1
T_w	Wash-out time constant (s)	10.0
K_{ac}	AC voltage droop (pu)	0.1
K_{fpod}	POD gain (pu)	200.0
$u_{ac,thr}$	AC voltage threshold (pu) for voltage control	10^{-6}
f_{thr}	Frequency threshold (pu) for POD	10^{-6}
ΔQ_{max}	Max. power variation (pu)	0.4
ΔQ_{min}	Min. power variation (pu)	-0.4
$Ramp_q$	Max. power ramp (pu/s)	1000
$u_{pod,thr}$	AC voltage threshold (pu) for POD	10^{-6}

Table 2.12: CON parameters

Parameter	Description	Typical value
ISVACCNTRL	Enable distributed DC control (0/1)	1
ISFCNTRL	Enable POD (0/1)	0

Table 2.13: ICON parameters

The state variables are the following:

State	Description
x_ω	Frequency lag filter
x_w	Wash-out filter

Table 2.14: STATE variables

WDELAY - Wide area measurements delays

The WDELAY mode implements the communication delays between n_{vsc} VSC stations for wide-area controls. The communication delays can be constant or they can vary randomly according to a triangular density function with deviation, *sigma*. WDELAY is implemented as a minimum excitation limiter model.

The following format is used in the .dyr file to assign and parameterize the WEDLAY model:

BUS 'USRMDL' IM 'WDELAY' IC IT NI NC NS NV ICON(M) ICON(M+n) CON(J) to CON(J+n)

where

- BUS: the number of the AC bus f of the VSC of the ith MTDC system
- IM: the bus ID of the VSC generator according to main_hyacdcsim (e.g., '14').
- IC: 9
- IT: 0
- NI: 2+n_{vsc}
- NC: 1+n_{vsc}
- NS: $2 \cdot n_{vsc}$
- NV: 1+n_{vsc}

The ICON and CON parameters are as follows:

Parameter	**Description**	**Typical value**
$\boldsymbol{\tau}$	Median delay for each VSC ($\in \mathbb{R}^{n\times 1}$)	-
σ	Deviation of the triangular density function	0.0

Table 2.15: CON parameters

Parameter	Description	Typical value
MONOFF	Enable delays (0/1)	0
n_{vsc}	Number of VSCs	-
BUS	AC bus numbers of the n_{vsc} VSCs ($\in \mathbb{Z}^{n_{vsc} \times 1}$)	

Table 2.16: ICON parameters

The state variables are the following:

State	Description
$\boldsymbol{x_{p,1}}$	First state variables of second-order Padé approximation ($\in \mathbb{R}^{n_{vsc} \times 1}$)
$\boldsymbol{x_{p,2}}$	Second state variables of second-order Padé approximation ($\in \mathbb{R}^{n_{vsc} \times 1}$)

Table 2.17: STATE variables

2.4.3 Dynamic Case Study

The dynamic performance of the proposed AC/DC model is evaluated using the same modified Kundur system introduced in the steady-state case study.

Two different representations of the DC grids are considered. In the first case, the dynamic DC-grid model, DDCGRD, is employed, in which the DC-bus voltages and DC-line currents are represented by dynamic state variables. In the second case, the static DC-grid model, SDCGRD, is used, in which the DC networks are solved algebraically without introducing dynamic state variables for the DC grids.

The same AC system, converter configuration, operating point, and two independent DC grids are considered in both simulations. The VSCGFL model is used to represent GFol VSC converters. Therefore, the obtained responses allow the influence of the DC-grid representation on the overall dynamic behavior of the interconnected AC/DC system to be assessed under identical operating and disturbance conditions.

The parameters used for the dynamic simulations are summarized in Table 2.18.

Parameter	**Value**
Simulation time step, T_{step}	0.001 s
Acceleration factor, TYSLACCFAC	0.9
Solution tolerance, TYSLTOL	1×10^{-4}
Maximum number of iterations, TYSLMAXITER	75
Disturbance application time	1.1 s
Final simulation time, T_{final}	12 s

Table 2.18: Parameters used for the dynamic simulations.

The dynamic simulation is initiated at $t = 0$ s, and the system is first simulated under its initial operating conditions until $t = 1.1$ s. At this instant, a disturbance is introduced by changing the generator reference (GREF) of the machine with ID 1 connected to AC bus 3 to 0.65. The disturbance is applied through the Python-based automation script used to control the PSS/E simulation, using the following command:

```
psspy.change_gref(3, r"1", 0.65)
```

After applying the disturbance, the simulation is continued until $t = 12$ s to evaluate the post-disturbance response of the AC/DC system.

The dynamic response obtained using the DDCGRD representation is shown in Figure 2.6. In this case, the DC-grid dynamics are explicitly represented through the DC-bus voltage and DC-line current state variables.

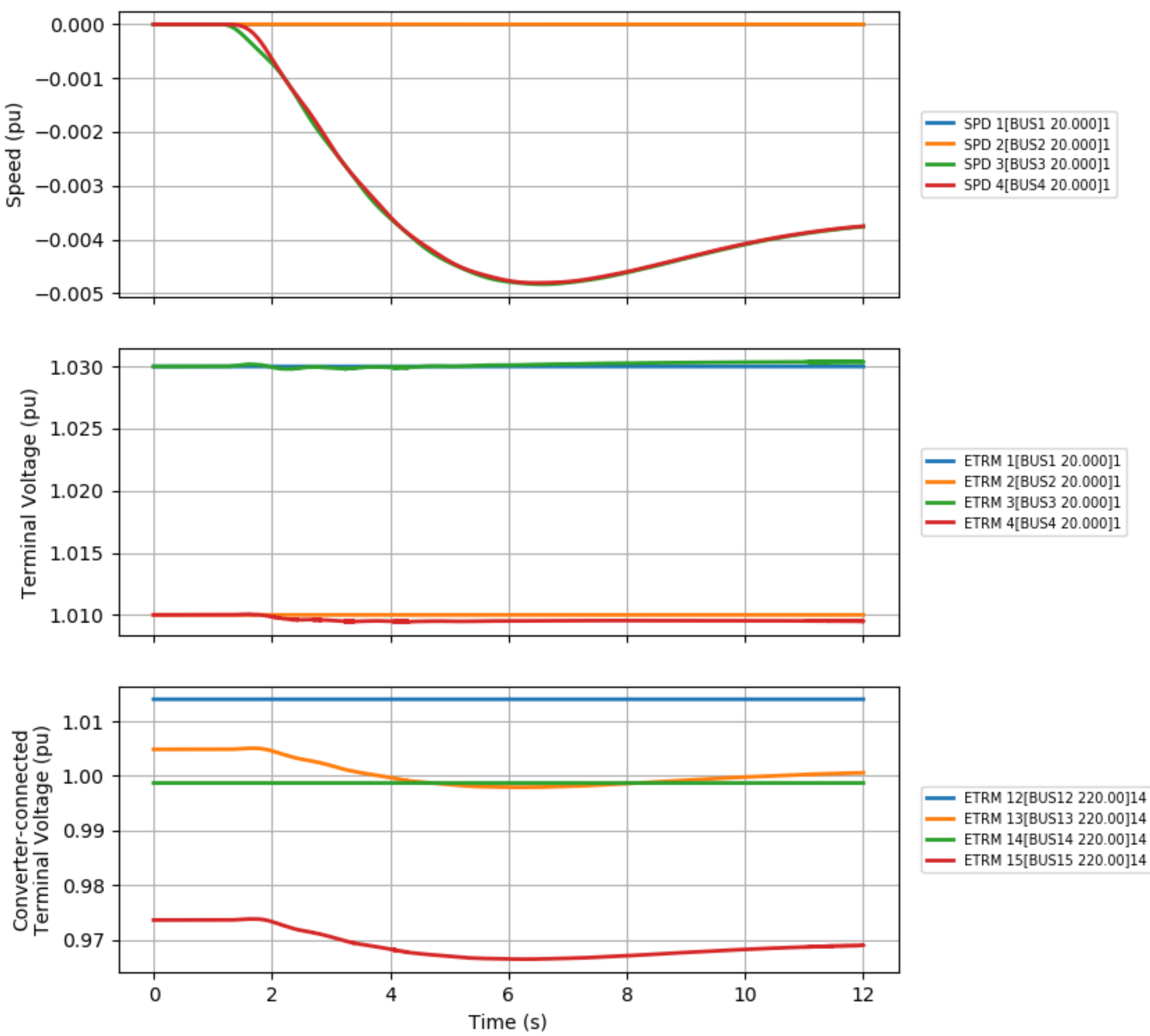


Figure 2.6: Dynamic response of the AC/DC system using the DDCGRD model.

The corresponding dynamic response obtained using the SDCGRD representation is shown in Figure 2.7. In this case, the DC networks are represented algebraically and no dynamic state variables are introduced for the DC grids.

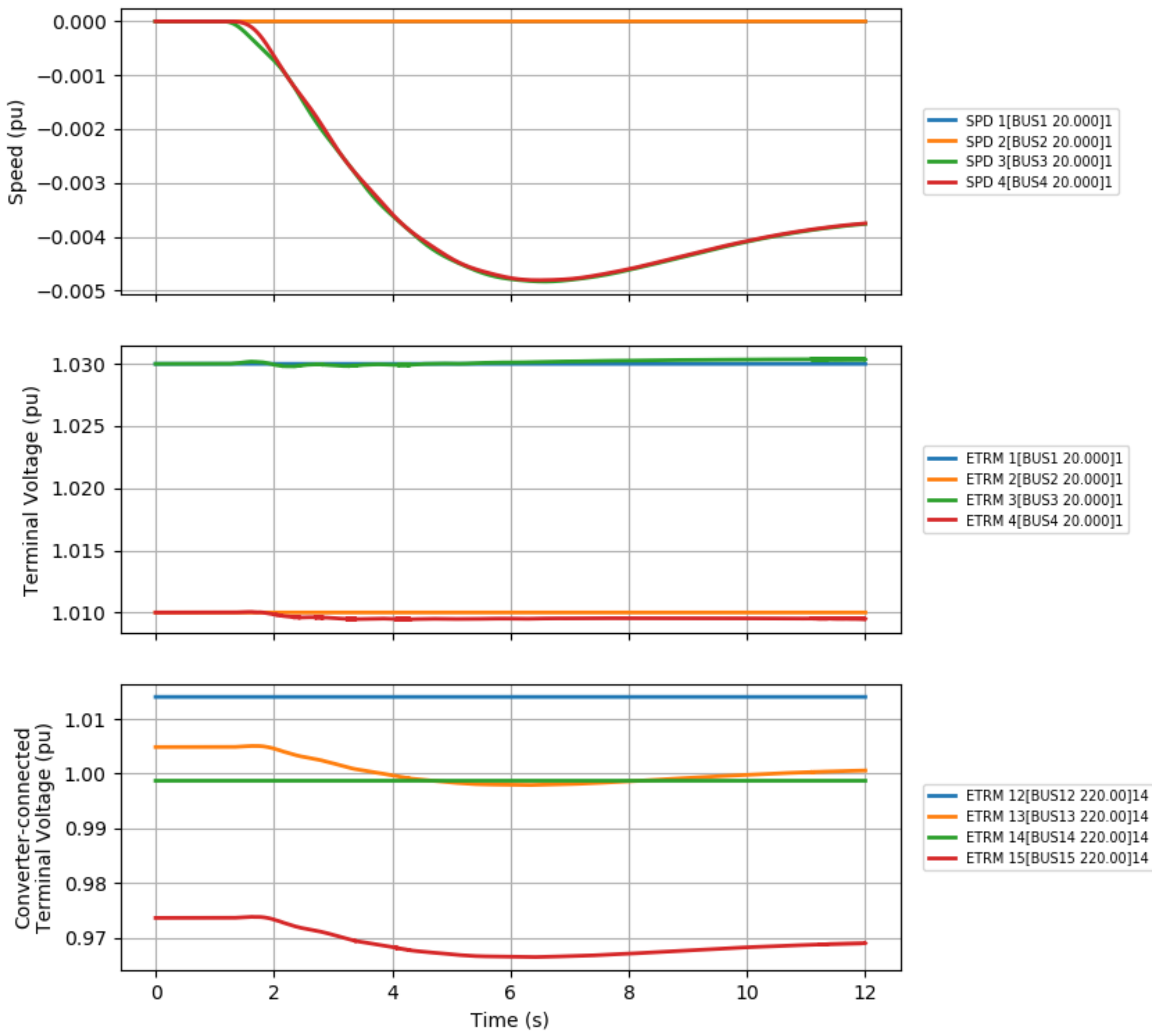


Figure 2.7: Dynamic response of the AC/DC system using the SDCGRD model.

The same set of variables is presented in both figures to enable a direct comparison between the two DC-grid representations. The upper subplot shows the speed responses of the synchronous machines with machine ID 1. The middle subplot presents the terminal-voltage magnitudes of the same machines. The lower subplot shows the terminal-voltage magnitudes associated with the VSC representations, corresponding to the machine models with ID 14 connected to the virtual AC buses.

Following the change in the generator reference at $t = 1.1$ s, deviations are observed in the machine speeds and terminal-voltage magnitudes. The responses remain bounded throughout the simulated time interval. The comparison between Figures 2.6 and 2.7 illustrates the effect of representing the DC grids dynamically or algebraically on the transient response of the overall AC/DC system.

The computational performance of the two DC-grid representations is also compared under the same simulation conditions. The execution time is approximately 5.546 s when the DDCGRD model is employed and approximately 5.145 s when the SDCGRD model is used. Thus, for the considered case study, the static DC-grid representation requires slightly less computational time than the dynamic representation. This reduction is consistent with the fact that the SDCGRD formulation does not introduce DC-grid dynamic state variables. The reduction is further only small since the same simulation time step of 0.001 s has been used. Nevertheless, the reported execution times are specific to the present case study and computational setup and should not be interpreted as general performance characteristics of the two formulations.

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